



University of
New Haven

TAGLIATELA
COLLEGE OF ENGINEERING

**EDU PREDICT TOOL
PREDICTING ENROLLMENT TRENDS IN
HIGHER EDUCATION IN THE USA**

Project Details:



Project Title: EDU PREDICT TOOL



Project Team: 01



Project Advisor: Dr. Ardiana Sula



MSDS Capstone – SP 25 DSCI 6051-07



Date of Presentation: 02-28-2025

EduPredict – Forecasting Higher Education Enrollment Trends

- **Goal:** Analyze and predict future enrollment patterns for international students in U.S. universities using data-driven techniques.
- **Outcome:** A **Power BI-based interactive dashboard** that provides enrollment trend forecasts under different scenarios (**Baseline, Growth, and Decline**).
- **Core Features:**
 - Machine learning-driven insights for **strategic academic planning**.
 - Customizable filters for **region, study level, and time period** to explore data dynamically.
- **Significance:**
 - Equips university officials with insights to **optimize resource planning and policy decisions**.
 - Supports institutions in **adapting to demographic and economic shifts** in student enrollment.

TEAM



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The Problem

1

Unpredictable Enrollment Trends

- ❖ Universities struggle to accurately forecast student enrollments due to shifting demographics, changing student preferences, and external influences.

2

Impact of External Factors

- ❖ Enrollment is heavily influenced by **government policies, economic conditions, and global mobility trends**, making long-term planning difficult.

3

Data Fragmentation & Decision-Making Challenges

- ❖ Enrollment data is scattered across multiple sources, leading to incomplete insights and reactive rather than proactive strategic planning.

Project Goals & Objectives



Primary Goal: Develop a **predictive model** that estimates future **student enrollments** based on multiple factors.



Key Objectives:



Integrate **historical datasets** on enrollment, demographics, and funding sources.



Identify **patterns & trends** in student admissions across various academic levels.



Build a **forecasting model** using machine learning.



Deploy results via a **user-friendly Power BI dashboard** for dynamic analysis.

Data Collection & Sources

Title: Where Our Data Comes From

Content:

- **Dataset Breakdown:**
 - **Student Enrollment Trends** (By year, region, academic level).
 - **Demographics** (Gender, marital status, visa type).
 - **Funding Sources** (Self-funded, government-sponsored, institution grants).
- **Data Quality:**
 - Missing data handled using **forward fill & interpolation** techniques.
 - Cleaned, structured, and prepared for **model training**.

Exploratory Data Analysis (EDA)

Title: Making Sense of the Data

Content:

- **What We Did:**

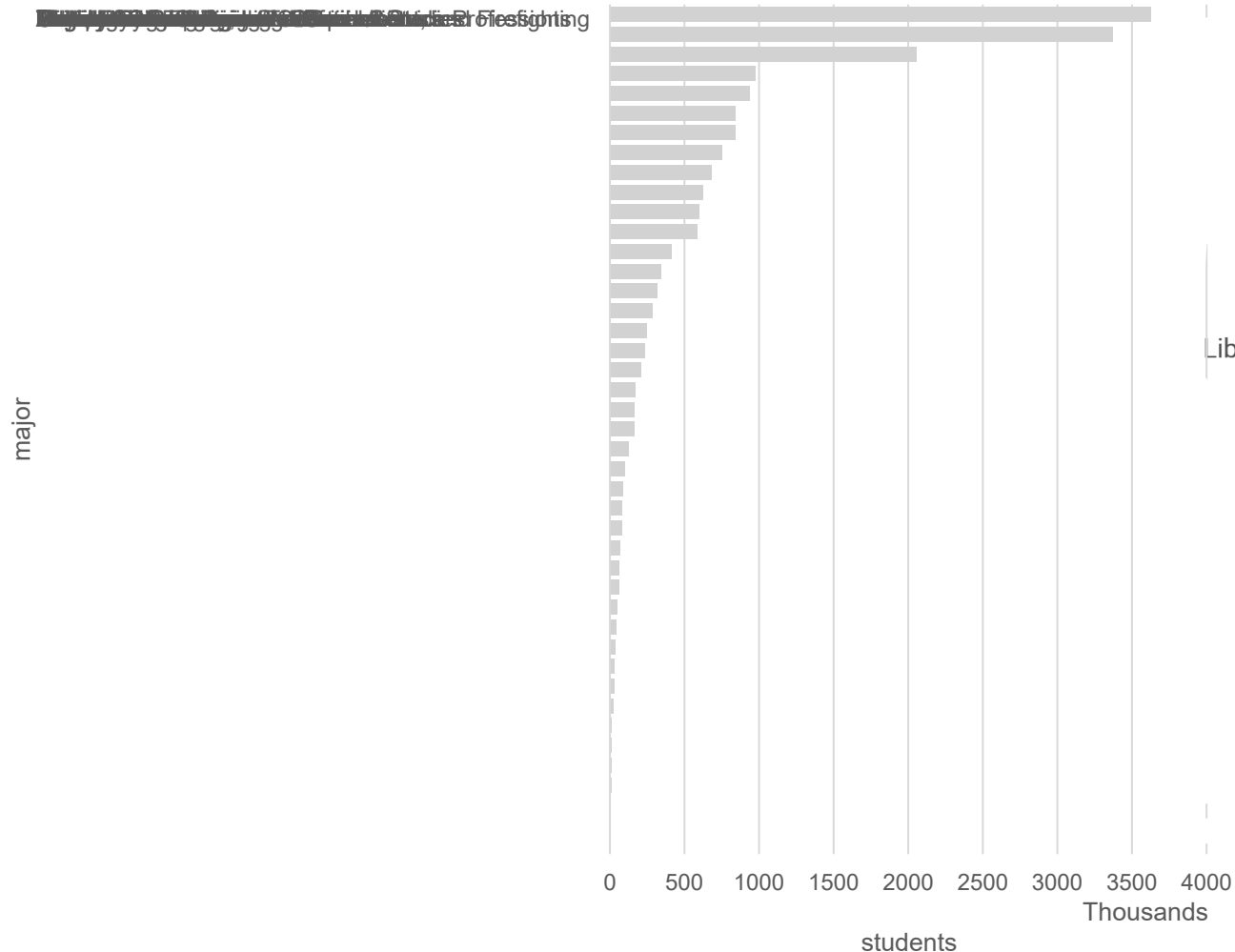
- Examined trends in **student enrollment across different years**.
- Analyzed **regional student distribution** to identify dominant source countries.
- Explored funding types to understand **how students finance their studies**.
- Correlation analysis to detect **relationships between key factors**.

- **Key Insights:**

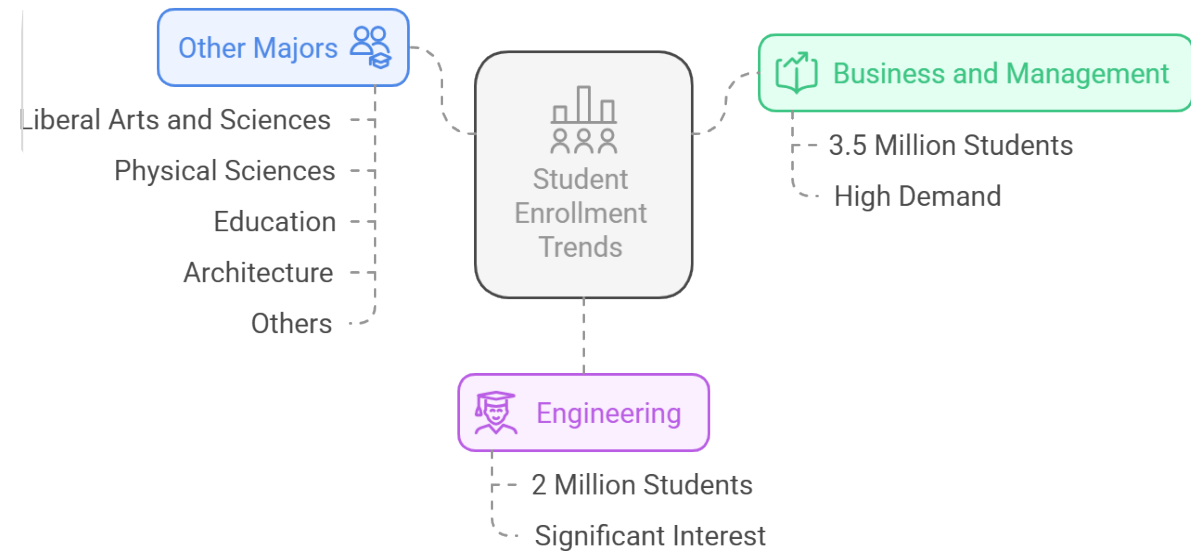
- **Business & STEM fields** are the most popular.
- **Asia contributes the highest number** of international students.
- Enrollment growth has **gradually increased over the years**.

Key EDA Visualizations

'major': **Business and Management** and **Engineering** have noticeably higher 'students'.

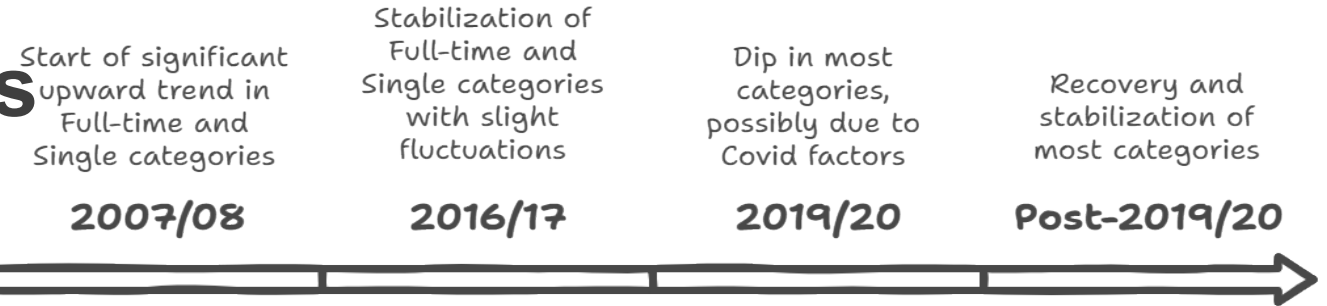


Student Enrollment Trends in Higher Education

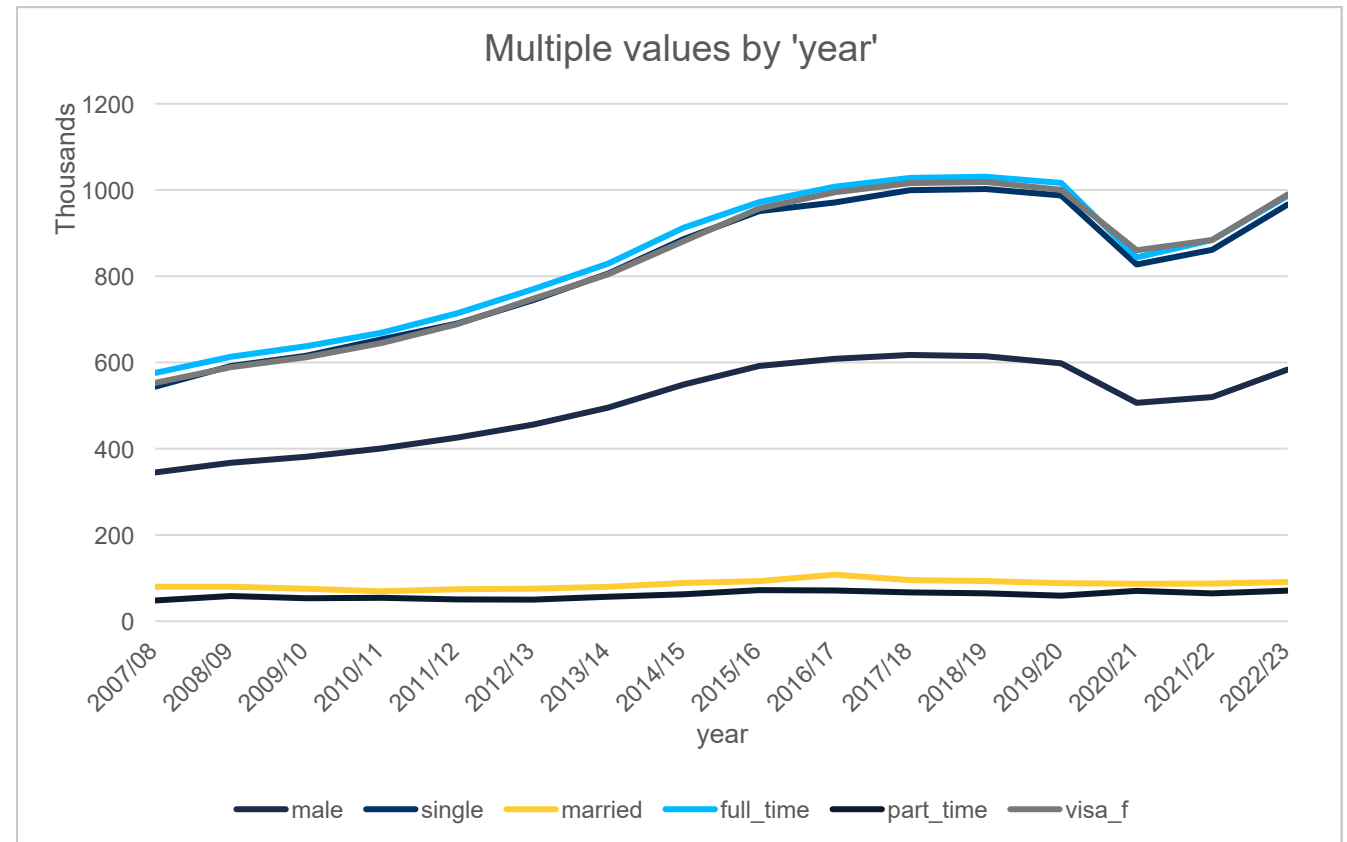
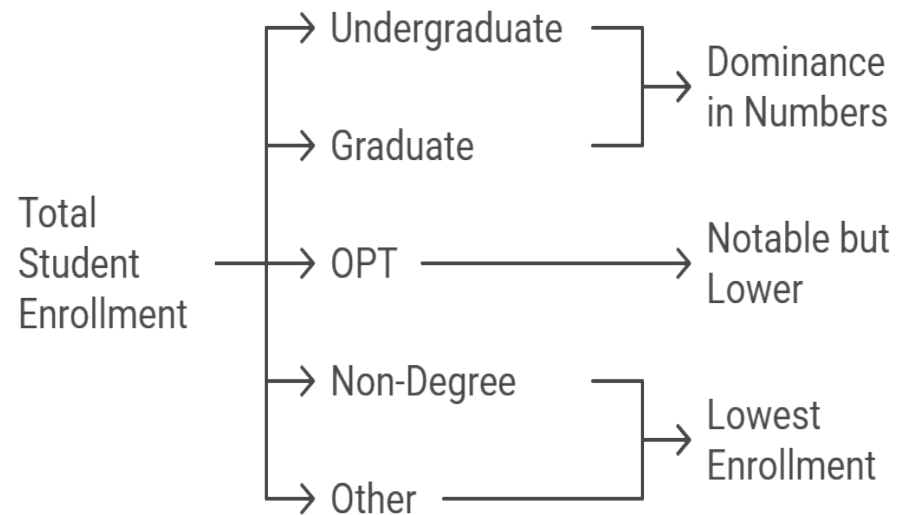


Key EDA Visualizations

Trends in Demographic Categories Over Time

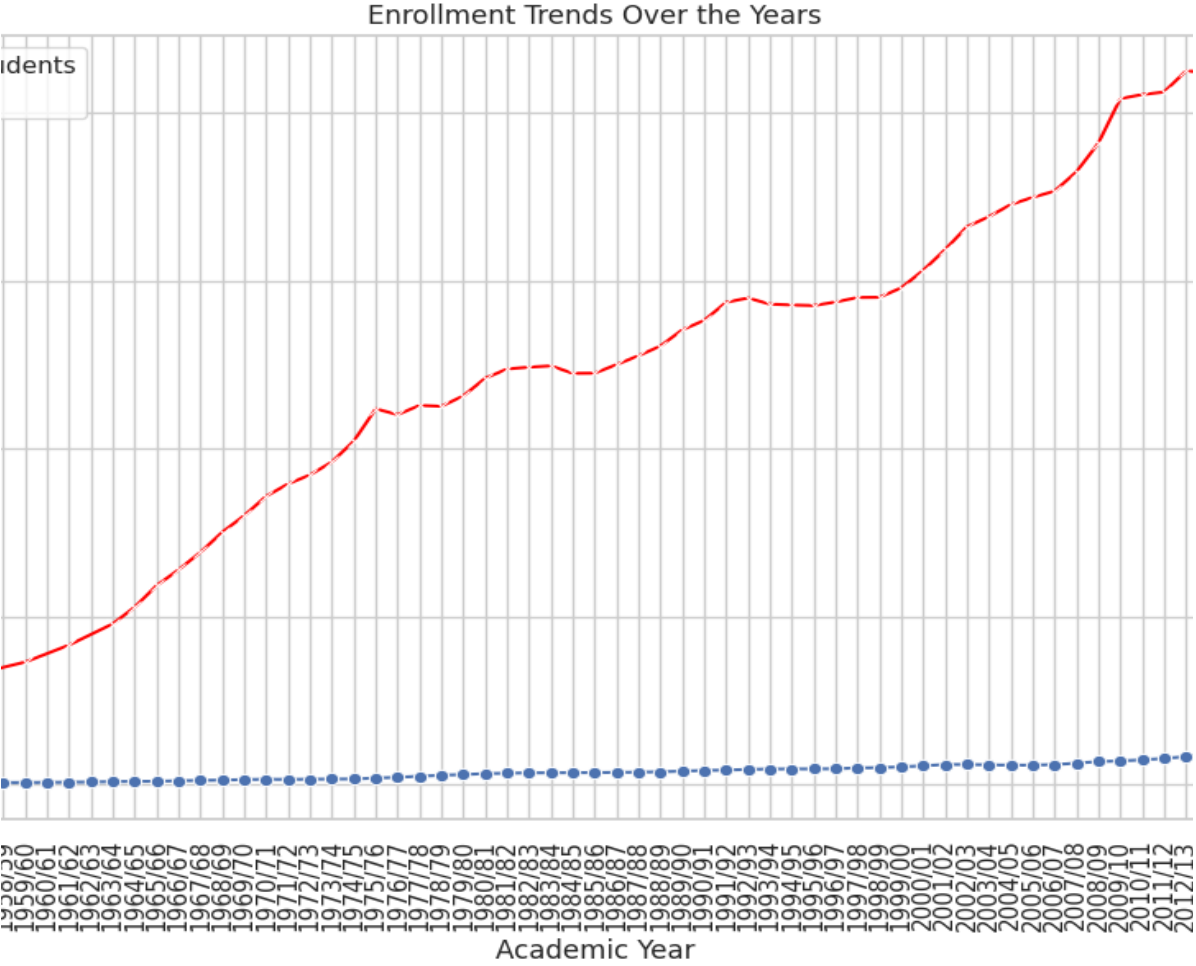


Student Enrollment Trends and Patterns

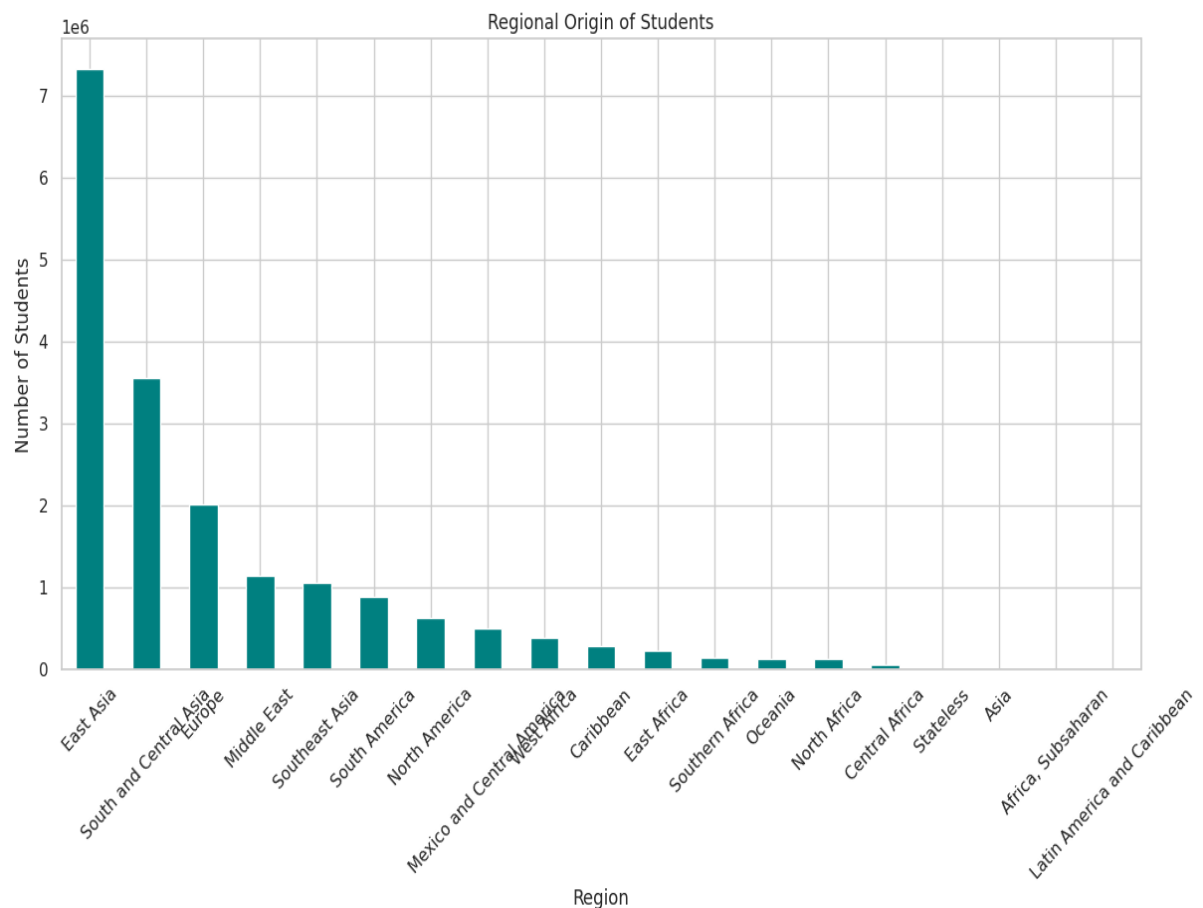


Key EDA Visualizations

Trend Analysis: Growth in **international vs. U.S. students** over time.

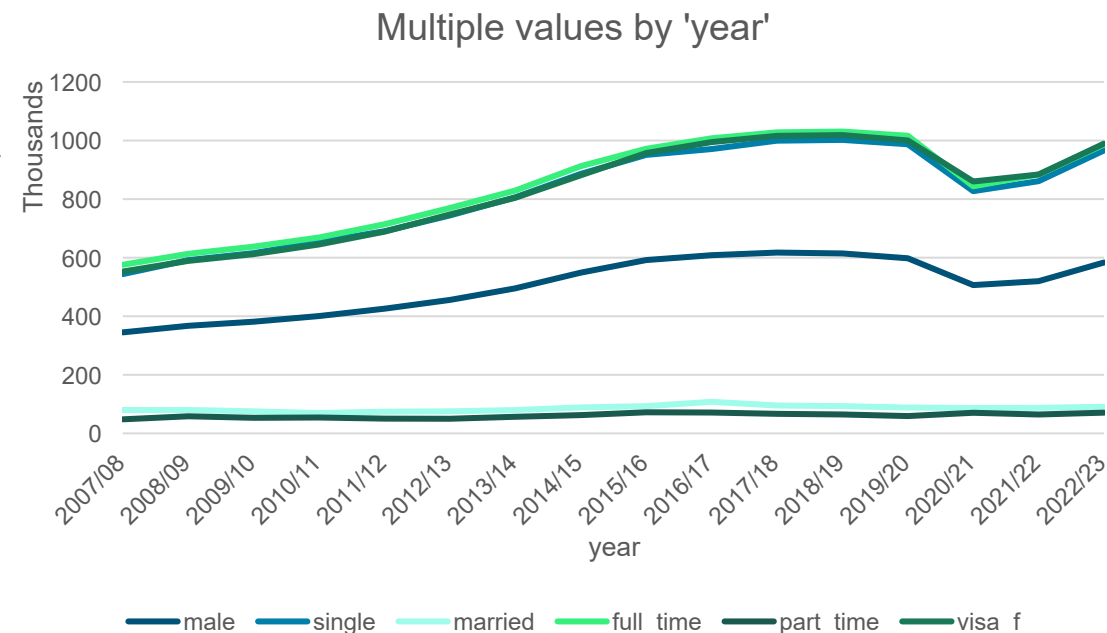
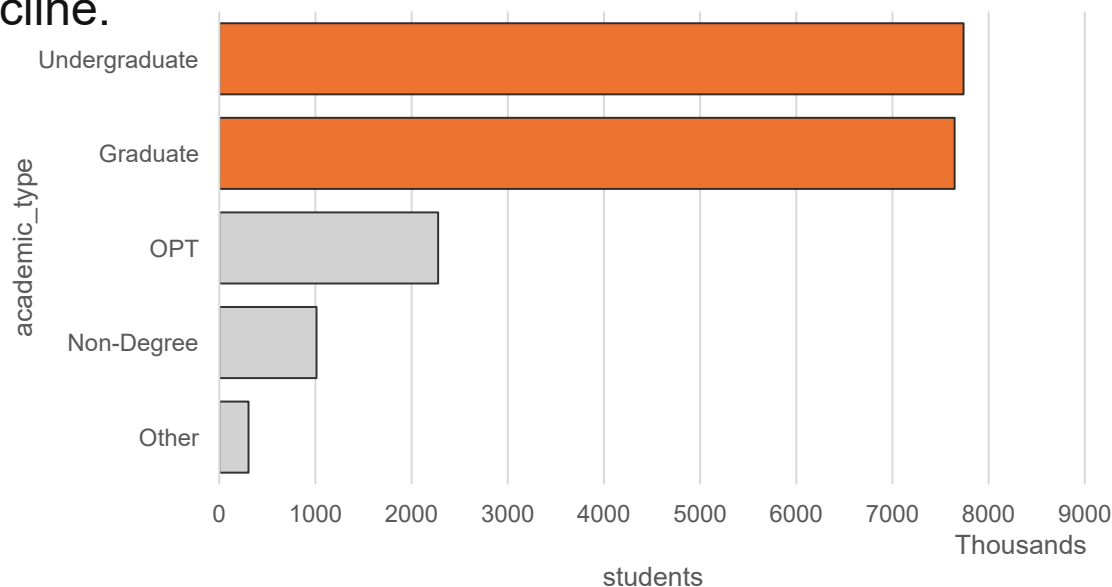


Regional Breakdown: Top contributing countries & continents for international students.



Key EDA Visualizations

- 1.Full-time, Married Increase:** These categories show upward trends.
- 2.Male, Part-time, Visa_f Stable:** Little to no change over time.
- 3.Single Declines:** Peaks around 2017/18, then decreases.
- 4.Overall Growth Until 2019/20:** Followed by slight decline.



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Trends in Employment Categories

Full-time Growth

Significant upward trend observed from 2007/08 to 2016/17.



Male Growth

Slower upward trend compared to full-time and single categories.



Married Stability

Remains relatively flat throughout the observed period.



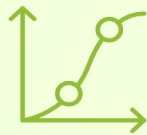
Observations Summary

Highlights notable growth and external factors affecting trends.



Single Growth

Notable increase from 2007/08 to 2016/17, stabilizing afterward.



Part-time Growth

Shows an upward trend but at a slower rate.



Visa_f Increase

Slight increase over the years, lowest among all categories.



Handling Missing Data

Title: Ensuring Data Accuracy

Content:

- **Challenges:**
 - Some datasets had missing values, particularly in **older academic years**.
- **Solutions Implemented:**
 - **Forward Fill:** Used to maintain **continuity in time-series data**.
 - **Linear Interpolation:** Applied to estimate missing records based on known values.
 - **Outlier Detection:** Cleaned inconsistent data to avoid skewing results.
- **Outcome:**
 - A well-structured, **high-quality dataset** for training predictive models.

Modeling Approach

Title: Choosing the Right Prediction Models

Content:

- **Forecasting Techniques Used:**

- **ARIMA:** For traditional time-series forecasting.
- **Prophet:** Captures seasonality and event-driven changes.
- **XGBoost:** Handles complex interactions in **demographic and financial factors**.

- **Trade-Offs:**

- ARIMA works well for **time-dependent trends** but lacks flexibility.
- Prophet helps with **seasonal patterns** but needs additional tuning.
- XGBoost is **highly accurate** but computationally expensive.

Baseline & Progress [02-26-2025 to 03-02-2025]

Title: What We Have Done So Far

Content:

- **Reviewed past submissions** and finalized project scope.
- **Conducted detailed EDA** to understand **key trends & data gaps**.
- **Handled missing data** using **forward-fill and interpolation**.
- **Developed the first baseline forecasting model (ARIMA)** to test initial predictions.
- **Refined feature selection** to improve model accuracy.

Bi-Weekly Key Milestones Report

Title: Tracking Our Progress

Content:

- **Week 1 Progress:**
 - Reviewed **project scope and data sources**.
 - Conducted **data cleaning & preprocessing**.
 - Built **initial visualization reports** for trend analysis.
- **Next Steps:**
 - Enhance **model accuracy** using hyperparameter tuning.
 - Explore **alternative models** for better forecasting.
 - Start integrating insights into **Power BI dashboard**.

Implementation & Deployment

Feature Engineering:

Categorical encoding,
normalizing
numerical features.

Model Optimization:

Hyperparameter
tuning to **increase
predictive accuracy.**

Deployment

Strategy: Integrating
forecasts into an
**interactive Power BI
dashboard &
Enabling user-driven
filtering** for
customized trend
analysis.

Next steps



Refine our EDA to enhance trend interpretation.



Improve model accuracy with fine-tuned parameters.



Test different machine learning models for better performance.



Build a user-friendly dashboard for interactive data visualization.



Document insights for reporting and final project submission.

THANK YOU

Please let us know if you have any further suggestions or improvements.



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